TALAL MALIK

Automotive Engineering | Energy Harvesting | Regenerative Systems | Computational Modeling

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PROFESSIONAL SUMMARY

Automotive engineer (B.E., NED University of Engineering and Technology, 2025) with research experience in vehicle energy efficiency, regenerative braking, electromagnetic and thermoelectric energy harvesting, transmission kinematics, and predictive analytics. Co-author of an EngrXiv preprint on hybrid vehicle energy recovery and designer of a funded, prototyped regenerative braking and suspension system. Currently a Product Manager (Localization) at KIA Pakistan, applying engineering analysis and data-driven methods (Python, SQL, Power BI) to component development and supplier performance. Seeking a PhD or research-based Master's program in automotive, mechanical, or energy systems engineering.

EDUCATION

Bachelor of Engineering, Automotive Engineering **Graduated: 2025**
NED University of Engineering and Technology, Karachi, Pakistan | CGPA: 3.00 / 4.00

- **Final Year Project:** Design and prototype of a regenerative anti-lock electro-mechanical braking system with regenerative suspension (funded).
- **Relevant Coursework:** Modeling and Simulation, Control Systems, Mechatronics, Embedded Systems, Vehicle Instrumentation, Metallurgy, Applied Probability and Statistics, Product Design and Development, Quality Management, Computer Programming.

RESEARCH INTERESTS

Energy Systems and Energy Harvesting | Regenerative Braking and Suspension | Automotive Mechatronics | Materials and Metallurgy | Computational Modeling and Simulation | Data-Driven Engineering and Predictive Analytics

PUBLICATIONS AND MANUSCRIPTS

Advanced Vehicle Hybrid Energy Recovery: A Novel Suspension-Based Electromagnetic and Thermal Harvesting Coupled with Regenerative Braking. *Preprint, EngrXiv (Co-Author). DOI:* https://doi.org/10.31224/8360

Simulation and Analysis of Gear Kinematics for Automobile Transmission. *Manuscript in Preparation (Research Assistant).*

RESEARCH EXPERIENCE AND PROJECTS

Regenerative Anti-lock Braking and Suspension System (Final Year Project) **2025**
CATIA | SolidWorks | MATLAB | Simulink | ANSYS | Arduino | C++

- Designed and prototyped a hybrid automotive energy recovery system integrating regenerative braking, electromagnetic suspension, and thermoelectric energy harvesting to improve EV energy efficiency.
- Developed engineering simulations and a functional prototype; received Ignite Final Year Project funding (2025).

Gear Kinematics Simulation and Analysis for Automobile Transmission **Research Assistant**

- Conducted simulation and kinematic analysis of automotive transmission gear systems; manuscript in preparation.

Predictive Model for Vendor Performance **Co-Author, In Progress**
Python | Machine Learning | SQL | Power BI | Excel

- Developing a machine learning model to predict supplier performance and project delivery risk from quality, delivery, cost, and production data.

PROFESSIONAL EXPERIENCE

Product Manager, Localization **May 2026 – Present**
Lucky Motor Corporation (KIA Pakistan), Karachi, Pakistan

- Lead component localization across the Kia Sorento, Sportage, and Stonic platforms, coordinating engineering, R&D, quality, procurement, and supplier teams to turn product requirements into locally manufacturable components.
- Evaluate localization feasibility by analyzing engineering changes, compliance requirements, cost targets, and supplier capabilities.
- Apply Python, SQL, and Power BI to supplier quality, delivery, and performance data to identify patterns and improvement opportunities.
- Coordinate APQP/PPAP documentation, engineering change processes, and technical issue resolution across 30+ suppliers and internal stakeholders.

Lead Project Manager **April 2024 – April 2026**
Design District, Karachi, Pakistan

- Managed multidisciplinary teams across 200+ software and digital transformation projects, coordinating technical execution, resource allocation, and structured problem-solving to improve delivery efficiency.

Project Manager **March 2021 – February 2024**
Olive Digital, Karachi, Pakistan

- Raised client retention to approximately 95% and cut project revision cycles by 50% through requirements gathering, documentation, and stakeholder management.

ENGINEERING INTERNSHIPS

Orient Energy Systems **December 2024 – January 2025**

- Assisted in assembly, testing, and inspection of diesel generators up to 100 kVA.

Toyota Highway Motors **March 2024 – April 2024**

- Supported workshop operations, inventory monitoring, and quality initiatives following Toyota Production System (TPS) practices.

TECHNICAL SKILLS

Programming and Data Analytics: Python, SQL, Machine Learning, C++, Power BI, Git, Excel, Minitab

CAD and Simulation: CATIA, SolidWorks, Fusion 360, ANSYS, MATLAB, Simulink

Manufacturing and Quality: APQP, PPAP, FMEA, Control Plans, SPC, Lean Manufacturing, Kaizen, Kanban, TPS, 5S, GD&T, Root Cause Analysis, Engineering Change Management (ECN/ECR), BOM Management, Supplier Development, Cost Reduction

Project and Enterprise Tools: SAP MM and PP, Jira, ClickUp, Smartsheet, Trello, HubSpot

CERTIFICATIONS

- IBM Data Analyst Professional Certificate, IBM
- Master Quality Management: SPC, Lean and Six Sigma, EDUCBA
- CAD/CAM/CAE for Mechanical Engineering, Autodesk
- Programming for Everybody (Python), University of Michigan

AWARDS AND LEADERSHIP

- **Ignite Final Year Project Funding (2025):** Awarded funding for the final year project.
- **IMechE SOFE Competition:** Runner-up, university-wide presentation competition.
- **Propellair 2024, Team Leader:** Led the design and development of an RC aircraft.

LANGUAGES

English (Fluent) | Urdu (Native) | Turkish (Basic) | Arabic (Basic)